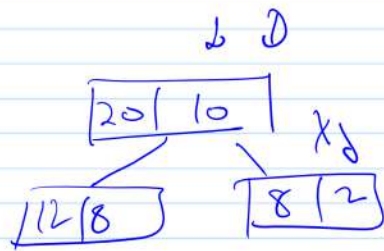


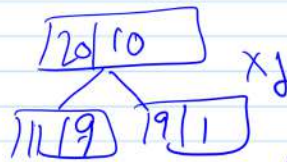
COL 794 / 7341

Oct 10, 2025



$x_D, x_{D'}$

D'



$$\{x^{(i)}, y^{(i)}\}_{i=1}^m$$

$$\{x^{(k)}, y^{(k)}\}_{k=2}^m$$

$$(x^{(i)}, y^{(i)}) \sim \text{Dist}$$

f: target function

Bias:-

$$\hat{M}(n) = E_{M: \{x^{(i)}, y^{(i)}\}_{i=1}^m} [M(n)]$$

$(x^{(i)}, y^{(i)}) \sim \text{Dist}$

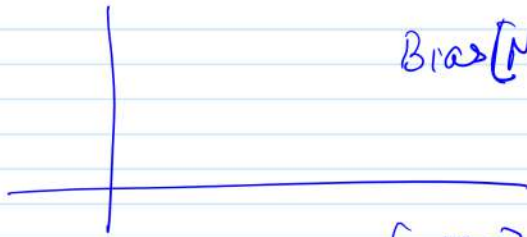
K -
subset

$$\{x^{(i)}, y^{(i)}\}_{i=1}^m$$

$$\{x^{(k)}, y^{(k)}\}_{k=2}^m$$

$$\{x^{(k)}, y^{(k)}\}_{k=2}^m$$

$$\text{Bias}[M(n)] = \|f(n) - \hat{M}(n)\|$$



$$\text{Var}[M(n)] = E \left[\|M(n) - \hat{M}(n)\|^2 \right]$$

$M: \{x^{(i)}, y^{(i)}\}_{i=1}^m$
 $(x^{(i)}, y^{(i)}) \sim \text{Dist}$